Networking Basics for JDE CNC Administrators

> *A practical guide covering foundational networking concepts mapped directly to JDE EnterpriseOne administration. Every topic includes real-world CNC scenarios, commands, and troubleshooting examples.*

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1. OSI Model & TCP/IP

What It Is

The OSI (Open Systems Interconnection) model breaks network communication into **7 layers**. TCP/IP condenses these into **4 practical layers** that real networks actually use. As a CNC admin, you use this model as a **diagnostic framework** — when something breaks, you work bottom-up to find where it failed.

```
| OSI Layer      | TCP/IP Layer   | What Happens Here                       |
| -------------- | -------------- | --------------------------------------- |
| 7 Application  | Application    | HTTP, HTTPS, DNS, LDAP - JDE UI, APIs   |
| 6 Presentation | Application    | SSL/TLS encryption - JDE certificates   |
| 5 Session      | Application    | JDENET sessions between servers         |
| 4 Transport    | Transport      | TCP connections - port 6009, 1521, 8080 |
| 3 Network      | Internet       | IP addresses, routing between servers   |
| 2 Data Link    | Network Access | Ethernet, MAC addresses, VLANs          |
| 1 Physical     | Network Access | Cables, NICs, switches                  |
```

The JDE Component Stack

```
User Browser
    |  HTTPS (Layer 7)
    v
JDE HTML / Web Server (WebLogic/Tomcat)   ← Layer 7: HTTP/HTTPS on port 8080/443
    |  JDENET TCP (Layer 4/5)
    v
JDE Enterprise Server                      ← Layer 4: TCP port 6009–6017
    |  JDBC TCP (Layer 4)
    v
Oracle / SQL Server Database               ← Layer 4: TCP port 1521 (Oracle)
```

Real-Time Example: User Cannot Log In to JDE

A user calls saying they cannot open the JDE login page. Here is how you diagnose using the OSI model — bottom-up:

Step 1 — Layer 3 (Network): Can the web server even be reached?

```
# From your machine, ping the JDE web server
ping jde-web-server01

# Expected: Reply from 10.10.1.50
# Bad: Request timed out  →  routing or firewall blocking ICMP
```

Step 2 — Layer 4 (Transport): Is the port open?

```
# Test if port 8080 is accepting connections
telnet jde-web-server01 8080

# Windows alternative
Test-NetConnection -ComputerName jde-web-server01 -Port 8080

# Expected: Connected to jde-web-server01
# Bad: Connection refused  →  WebLogic not started, or firewall blocking port
```

Step 3 — Layer 6 (Presentation): Is the SSL certificate valid?

```
# On the JDE web server, check certificate in the keystore
keytool -list -v -keystore /u01/jde/keystore.jks -storepass changeit | grep "Valid"

# Look for: Valid from: Mon Jan 01 ... until: Wed Dec 31 ...
# Bad: Expired date  →  renew and reimport the certificate
```

Step 4 — Layer 7 (Application): Is WebLogic actually serving pages?

```
# Check WebLogic server log on the JDE web server
tail -100 /u01/Oracle/Middleware/user_projects/domains/JDE_E1/servers/JDE_Server1/logs/
JDE_Server1.log

# Look for: <Server started in RUNNING mode>
# Bad: OutOfMemoryError, SocketException  →  JVM heap issue or JDENET misconfiguration
```

> ***CNC Rule:*** *Never jump straight to the JDE application log. If the network path is broken at Layer 3 or 4, application-level fixes will change nothing.*

---

2. IP Addressing & Subnets

What It Is

Every device on a network has an **IP address** — a unique numeric label. In JDE environments, all servers (web, enterprise, database, deployment) are identified by their IP addresses. Subnets define which machines are on the same local network segment.

IPv4 Basics

```
IP Address:   10.10.1.50
              |  |  | |
              |  |  | └── Host part (which machine on this subnet)
              |  |  └──── Subnet
              |  └─────── Network
              └──────────── Network class / range

Subnet Mask:  255.255.255.0   (also written as /24)
              means: first 24 bits = network, last 8 bits = host

Network:      10.10.1.0/24   → allows hosts 10.10.1.1 to 10.10.1.254
Broadcast:    10.10.1.255    → sends to all hosts on this subnet
```

Private IP Ranges (Always Used for JDE Servers)

```
| Range                          | CIDR | Common Use in JDE               |
| ------------------------------ | ---- | ------------------------------- |
| 10.0.0.0 – 10.255.255.255      | /8   | Large enterprise networks       |
| 172.16.0.0 – 172.31.255.255    | /12  | Medium-sized enterprise         |
| 192.168.0.0 – 192.168.255.255  | /16  | Small office / lab environments |
```

Typical JDE Server IP Layout

```
Subnet: 10.10.1.0/24  (Internal Application Zone)
----------------------------------------------------
10.10.1.10   →  JDE Deployment Server (DS)
10.10.1.20   →  JDE Enterprise Server (ES)
10.10.1.21   →  JDE Enterprise Server 2 (ES - failover)
10.10.1.30   →  Oracle Database Server (DB)

Subnet: 10.10.2.0/24  (DMZ – Web Tier)
---------------------------------------------
10.10.2.10   →  JDE HTML Web Server 1
10.10.2.11   →  JDE HTML Web Server 2
10.10.2.5    →  Load Balancer VIP (Virtual IP)

Subnet: 10.10.3.0/24  (Management Zone)
---------------------------------------------
10.10.3.50   →  CNC Admin workstation
10.10.3.51   →  Monitoring server
```

Real-Time Example: Adding a New Enterprise Server

Your company is adding a second Enterprise Server for load balancing. You need to verify IP settings before registering it in JDE.

Step 1 — Check the IP configuration on the new server

```
# Linux (RHEL/OEL – common for JDE Enterprise Servers)
ip addr show
# or older command:
ifconfig

# Expected output:
# inet 10.10.1.21/24 brd 10.10.1.255 scope global ens192

# Windows Server
ipconfig /all
# Look for: IPv4 Address: 10.10.1.21, Subnet Mask: 255.255.255.0
```

Step 2 — Confirm it can reach the database server (different subnet requires routing)

```
ping 10.10.1.30        # same subnet – should always work
ping 10.10.2.10        # different subnet – needs router/firewall rule
```

Step 3 — Calculate if two servers are on the same subnet

```
Server A: 10.10.1.20 /24
Server B: 10.10.1.21 /24
Same network? 10.10.1.x — YES, direct communication, no router needed.

Server A: 10.10.1.20 /24
Server C: 10.10.2.10 /24
Same network? 10.10.1.x vs 10.10.2.x — NO, traffic must go through a router/firewall.
```

Step 4 — Register the new server IP in jde.ini

```
; On the HTML Server's jde.ini, add/update the Enterprise Server entry
[JDENET_KERNEL_DEF2]
kType=2
kConnect=1
kPortNumber=6010
; Reference the server by hostname (DNS resolves to IP)
; Make sure DNS has an A record: jde-es02 → 10.10.1.21
```

> ***CNC Rule:*** *JDE servers must always use* ***static IP addresses****. If the Enterprise Server's IP changes (e.g., DHCP lease expires), all jde.ini files pointing to it will break. See Section 6 for DHCP vs static.*

---

3. DNS & Hostname Resolution

What It Is

DNS (Domain Name System) translates human-readable hostnames (like `jde-enterprise01`) into IP addresses (like `10.10.1.20`). JDE relies on DNS heavily — all server-to-server communication in jde.ini uses **hostnames**, not IP addresses directly.

How Resolution Works

```
JDE HTML Server needs to connect to "jde-enterprise01"
          |
          ▼
1. Check local hosts file first
   (C:\Windows\System32\drivers\etc\hosts  or  /etc/hosts)
          |
          ▼ (not found in hosts file)
2. Ask DNS server (configured in NIC settings)
   DNS Server: 10.10.0.10
          |
          ▼
3. DNS responds: jde-enterprise01 → 10.10.1.20
          |
          ▼
4. HTML Server connects to 10.10.1.20 on port 6009
```

Key DNS Record Types for JDE

```
| Record Type | Purpose                        | JDE Example
|
| ----------- | ------------------------------ | ---------------------------------------
------------- |
| A           | Hostname → IPv4 address        | jde-enterprise01 → 10.10.1.20
|
| CNAME       | Alias → another hostname       | jde-es → jde-enterprise01 (alias for l
oad balancer) |
| PTR         | IP → Hostname (reverse lookup) | 10.10.1.20 → jde-enterprise01
|
```

Real-Time Example: JDENET Connection Failure After Server Rename

The enterprise server was renamed from `jdeapp01` to `jde-enterprise01` during a server migration. After the rename, the HTML Server cannot connect to the Enterprise Server.

Step 1 — Check what hostname the HTML Server is using (from jde.ini)

```
; On the HTML Server, open:
; C:\JDEdwards\E920\ini\JDE.INI  (Windows)
; /JDEdwards/E920/ini/jde.ini    (Linux)

[JDENET_KERNEL_DEF1]
kType=2
kConnect=1
kPortNumber=6009
; This hostname must resolve correctly:
; (hostname is set elsewhere in the config — check the server name entries)
```

Step 2 — Test DNS resolution from the HTML Server

```
# Windows
nslookup jde-enterprise01
# Expected: Name: jde-enterprise01, Address: 10.10.1.20
# Bad: can't find jde-enterprise01: Non-existent domain

# Linux
nslookup jde-enterprise01
# or
dig jde-enterprise01 A

# Also test reverse lookup
nslookup 10.10.1.20
# Expected: name = jde-enterprise01.yourdomain.com
```

Step 3 — If DNS is wrong, test with a hosts file override

```
# Add a temporary entry to the hosts file to bypass DNS
# Windows: C:\Windows\System32\drivers\etc\hosts
# Linux:   /etc/hosts

10.10.1.20   jde-enterprise01   jde-enterprise01.company.com

# Save file, then test:
nslookup jde-enterprise01
# Should now resolve from hosts file
```

Step 4 — Verify the Enterprise Server's own hostname matches

```
# On the Enterprise Server itself
hostname
# Must return: jde-enterprise01
# If it returns the old name 'jdeapp01', the OS hostname was not updated

# Linux — update hostname permanently
hostnamectl set-hostname jde-enterprise01

# Windows
Rename-Computer -NewName "jde-enterprise01" -Restart
```

Step 5 — Check the Deployment Server's OCM (Object Configuration Manager)

```
In JDE Web Client → Server Administration Workbench (SAW)
→ Server Map → Verify server name matches the new hostname exactly
→ OCM mappings referencing old hostname must be updated
```

> ***CNC Rule:*** *In JDE,* ***hostnames must match everywhere****: jde.ini, OCM/Server Map in the database, the OS hostname, and DNS. A mismatch at any one of these four places will break connectivity.*

---

4. Ports & Protocols

What It Is

A **port** is a numbered logical endpoint on a server. When a service starts, it "listens" on a specific port. JDE has many components, and each uses specific ports for communication. A firewall that blocks even one required port will silently break a specific JDE function.

TCP vs UDP

```
TCP (Transmission Control Protocol)
- Connection-oriented: establishes a connection before sending data
- Guarantees delivery, order, and error checking
- Used by: ALL JDE inter-component communication
- Example: HTML Server → Enterprise Server via JDENET (TCP 6009)

UDP (User Datagram Protocol)
- Connectionless: fire-and-forget
- No guarantee of delivery
- Used by: DNS queries (UDP 53), some monitoring tools
- JDE itself does not use UDP for core communication
```

Complete JDE Port Reference

```
| Component / Service         | Port(s)     | Protocol | Direction
|
| --------------------------- | ----------- | -------- | ------------------------------
|
| JDE HTML/Web Server (HTTP)  | 8080        | TCP      | Users → Web Server
|
| JDE HTML/Web Server (HTTPS) | 443 / 8443  | TCP      | Users → Web Server
|
| WebLogic Admin Console      | 7001        | TCP      | CNC Admin → Web Server
|
| JDE AIS Server              | 8080 / 8443 | TCP      | Mobile/API → AIS Server
|
| JDE Orchestrator Framework  | 8080 / 8443 | TCP      | Integrations → Orchestrator
|
| JDENET (Enterprise Server)  | 6009        | TCP      | Web Server → Enterprise Server
|
| JDE Kernel ports            | 6010–6017   | TCP      | Web Server → Enterprise Server
|
| Oracle Database             | 1521        | TCP      | Enterprise Server → DB
|
| SQL Server Database         | 1433        | TCP      | Enterprise Server → DB
|
| JDE Deployment Server RPC   | 445 / 139   | TCP      | Deployment operations
|
| LDAP (Active Directory)     | 389         | TCP      | JDE → AD (authentication)
|
| LDAPS (AD over SSL)         | 636         | TCP      | JDE → AD (secure auth)
|
| SSH (Linux server admin)    | 22          | TCP      | CNC Admin → Linux servers
|
| RDP (Windows server admin)  | 3389        | TCP      | CNC Admin → Windows servers
|
```

Real-Time Example: Business Function (BSFN) Calls Hanging

A custom orchestration is calling a Business Function on the Enterprise Server. The calls hang for 60 seconds and then time out. Users report BSFNs are slow or failing.

Step 1 — Check if the JDENET port is open and responsive

```
# From the HTML/AIS Server, test connectivity to Enterprise Server JDENET port
telnet jde-enterprise01 6009
# Expected: blank screen (connected, waiting)
# Bad: 'Could not open connection' after timeout  →  port blocked or service down

# More informative test on Windows:
Test-NetConnection -ComputerName jde-enterprise01 -Port 6009
# TcpTestSucceeded: True  ← good
# TcpTestSucceeded: False ← port blocked or service not listening
```

Step 2 — Check listening ports on the Enterprise Server itself

```
# Windows Enterprise Server
netstat -an | findstr "6009"
# Expected: TCP  0.0.0.0:6009  0.0.0.0:0  LISTENING

# Linux Enterprise Server
netstat -tlnp | grep 6009
# or
ss -tlnp | grep 6009
# Expected: LISTEN  0  128  0.0.0.0:6009  0.0.0.0:*

# If port 6009 is NOT listed as LISTENING — the JDE kernel is not running
```

Step 3 — Check active JDENET connections (ESTABLISHED = connected clients)

```
# Windows
netstat -an | findstr "6009" | findstr "ESTABLISHED"

# Linux
netstat -an | grep 6009 | grep ESTABLISHED

# If you see ESTABLISHED connections, the service is up
# Count them — if MaxConnections is hit, new requests queue up → slowness
```

Step 4 — Check kernel port range in jde.ini

```
; On the Enterprise Server, open jde.ini
[JDENET]
serviceNameListen=6009
serviceNameConnect=6009

[JDENET_KERNEL_DEF1]
kType=2
kConnect=1
kPortNumber=6009      ; primary kernel port

; If you have multiple kernels, they use 6010, 6011, etc.
; Ensure ALL ports in this range are open in the firewall
```

> ***CNC Rule:*** *When you open a ticket with the network/firewall team, always give them the* ***exact source IP, destination IP, destination port, and protocol*** *(TCP/UDP). Example: "Please open TCP port 6009 from subnet 10.10.2.0/24 (web DMZ) to 10.10.1.20 (Enterprise Server)."*

5. Firewalls & ACLs

What It Is

A **firewall** is a security system that controls which network traffic is allowed to pass between network segments. Rules are evaluated in order — first match wins. An **ACL (Access Control List)** is the list of rules defining what is allowed or denied. In JDE environments, firewalls sit between the DMZ (web tier) and the internal network (application and database tiers).

How Firewall Rules Work

```
Rule evaluation (top to bottom, first match wins):

| Rule # | Source IP    | Destination IP | Port | Protocol | Action
|
| ------ | ------------ | -------------- | ---- | -------- | --------------------------
---------- |
| 1      | 10.10.2.0/24 | 10.10.1.20     | 6009 | TCP      | ALLOW   ← Web→ES JDENET
|
| 2      | 10.10.2.0/24 | 10.10.1.20     | 6010 | TCP      | ALLOW   ← Web→ES kernel
|
| 3      | 10.10.2.0/24 | 10.10.1.30     | 1521 | TCP      | DENY    ← Web cannot hit D
B directly |
| 4      | 10.10.1.20   | 10.10.1.30     | 1521 | TCP      | ALLOW   ← ES→DB Oracle
|
| 5      | ANY          | ANY            | ANY  | ANY      | DENY    ← Default deny all
|
```

JDE Three-Tier Firewall Architecture

```
Internet / Users
      |
      | HTTPS 443 / HTTP 8080
      ▼
 ┌──────────────────────────────────┐
 |         DMZ (Web Zone)           |
 |   JDE HTML Web Server            |
 |   JDE AIS Server                 |
 |   Load Balancer                  |
 └──────────────────────────────────┘
      |
      | JDENET TCP 6009-6017 (ALLOWED through firewall)
      | LDAP TCP 389 (if AD is in internal zone)
      ▼
 ┌──────────────────────────────────┐
 |      Internal (App Zone)         |
 |   JDE Enterprise Server          |
 |   JDE Deployment Server          |
 └──────────────────────────────────┘
      |
      | Oracle JDBC TCP 1521 (ALLOWED)
      ▼
 ┌──────────────────────────────────┐
 |      Restricted (DB Zone)        |
 |   Oracle / SQL Server            |
 └──────────────────────────────────┘

KEY RULE: Web Zone servers can NEVER directly reach the DB Zone.
          Only the Enterprise Server (App Zone) can talk to the database.
```

Real-Time Example: New JDE Environment — Firewall Change Request

You are deploying a new JDE EnterpriseOne 9.2 environment. You need to submit a firewall change request to the network security team.

Firewall Change Request (FCR) — what to include:

```
FCR Title: JDE EnterpriseOne 9.2 Production Deployment — Firewall Rules

Rules Required:
---------------------------------------------------------------------------
Rule 1: Web → Enterprise Server (JDENET)
  Source:      10.10.2.0/24  (DMZ Web Zone)
  Destination: 10.10.1.20    (jde-enterprise01)
  Port:        6009–6017
  Protocol:    TCP
  Direction:   Outbound from DMZ / Inbound to App Zone
  Purpose:     JDENET communication between HTML server and Enterprise Server

Rule 2: Enterprise Server → Oracle Database (JDBC)
  Source:      10.10.1.20    (jde-enterprise01)
  Destination: 10.10.1.30    (jde-oracle-db01)
  Port:        1521
  Protocol:    TCP
  Direction:   Outbound from App Zone / Inbound to DB Zone
  Purpose:     JDBC database connectivity for JDE business functions

Rule 3: CNC Admin → All JDE Servers (Administration)
  Source:      10.10.3.50    (cnc-admin-workstation)
  Destination: 10.10.1.20, 10.10.1.10, 10.10.2.10
  Ports:       22 (SSH), 3389 (RDP), 7001 (WebLogic console)
  Protocol:    TCP
  Purpose:     CNC administrative access to all servers

Rule 4: JDE → Active Directory (Authentication)
  Source:      10.10.1.20    (jde-enterprise01)
  Destination: 10.10.0.5     (ad-server01)
  Port:        389 (LDAP) or 636 (LDAPS)
  Protocol:    TCP
  Purpose:     JDE user authentication via Active Directory
---------------------------------------------------------------------------
```

Testing firewall rules after they are applied:

```
# Test each rule from the correct source machine

# Rule 1 — From the web server, test JDENET port:
Test-NetConnection -ComputerName 10.10.1.20 -Port 6009

# Rule 2 — From the Enterprise Server, test Oracle port:
telnet 10.10.1.30 1521

# Rule 4 — From the Enterprise Server, test LDAP:
Test-NetConnection -ComputerName 10.10.0.5 -Port 389
```

Diagnosing a blocked port (firewall rule missing):

```
# Symptom: JDE login fails with "Server is unavailable"
# Test from HTML Server:
Test-NetConnection -ComputerName jde-enterprise01 -Port 6009

# Output:
# ComputerName     : jde-enterprise01
# TcpTestSucceeded : False   ← firewall is blocking this port

# Escalate to network team:
# "TCP port 6009 from 10.10.2.10 to 10.10.1.20 is being blocked.
#  Please check ACL rule between DMZ and App Zone."
```

> ***CNC Rule:*** *Always test firewall rules* ***from the actual source server****, not from your admin workstation. A rule that allows 10.10.2.10 → 10.10.1.20 on port 6009 does not help you if you test it from 10.10.3.50 (your workstation).*

---

6. DHCP vs Static IP

What It Is

DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses to devices from a pool, with a time-limited "lease." When the lease expires, the device may get a different IP. **Static IP** means the IP address is manually configured and never changes.

Why JDE Servers Must Always Use Static IPs

```
Scenario: JDE Enterprise Server uses DHCP

Day 1:   Server gets IP  10.10.1.20 (DHCP lease)
         jde.ini on all HTML servers points to: jde-enterprise01 (DNS → 10.10.1.20)
         Everything works.

Day 30:  DHCP lease expires. Server reboots.
         Server gets new IP: 10.10.1.45 (new DHCP lease)
         DNS still points jde-enterprise01 → 10.10.1.20 (old IP)
         10.10.1.20 is now a DIFFERENT server or empty.

Result:  ALL JDE HTML servers cannot connect to Enterprise Server.
         JDENET connections fail. All users locked out.
         Firewall rules (written for 10.10.1.20) no longer apply.
         You chase a phantom problem for hours.
```

Static IP Assignment — JDE Server Checklist

Every JDE server needs these configured statically:

```
1. IP Address       →  e.g. 10.10.1.20
2. Subnet Mask      →  e.g. 255.255.255.0
3. Default Gateway  →  e.g. 10.10.1.1  (router that connects to other subnets)
4. DNS Server(s)    →  e.g. 10.10.0.10 (primary), 10.10.0.11 (secondary)
5. Hostname         →  e.g. jde-enterprise01 (must match DNS A record)
```

Real-Time Example: Configuring Static IP on a New JDE Enterprise Server

Linux (RHEL/Oracle Enterprise Linux) — most common for JDE Enterprise Server:

```
# Step 1 – Find the network interface name
ip addr show
# Look for interface like: ens192, eth0, enp3s0

# Step 2 – Edit the network configuration file
# RHEL 7/OEL 7:
vi /etc/sysconfig/network-scripts/ifcfg-ens192

# Set the following:
BOOTPROTO=none        # 'none' or 'static' – NOT 'dhcp'
IPADDR=10.10.1.20
NETMASK=255.255.255.0
GATEWAY=10.10.1.1
DNS1=10.10.0.10
DNS2=10.10.0.11
ONBOOT=yes
NAME=ens192

# Step 3 – Restart networking
systemctl restart network
# RHEL 8+:
nmcli connection reload
nmcli connection up ens192

# Step 4 – Verify
ip addr show ens192
# Expected: inet 10.10.1.20/24

# Step 5 – Test gateway reachability
ping 10.10.1.1
# Expected: Reply from 10.10.1.1
```

Windows Server — common for JDE Deployment Server and HTML Server:

```
# PowerShell — set static IP on a Windows JDE server

# Step 1 — Find interface index
Get-NetAdapter
# Note the InterfaceIndex, e.g. 6

# Step 2 — Remove existing DHCP assignment
Remove-NetIPAddress -InterfaceIndex 6 -Confirm:$false
Remove-NetRoute -InterfaceIndex 6 -Confirm:$false

# Step 3 — Set static IP
New-NetIPAddress `
  -InterfaceIndex 6 `
  -IPAddress 10.10.1.10 `
  -PrefixLength 24 `
  -DefaultGateway 10.10.1.1

# Step 4 — Set DNS servers
Set-DnsClientServerAddress `
  -InterfaceIndex 6 `
  -ServerAddresses 10.10.0.10, 10.10.0.11

# Step 5 — Verify
ipconfig /all
# Look for: IPv4 Address: 10.10.1.10, DHCP Enabled: No
```

DHCP Reservation — an Alternative for Non-Critical Servers

For dev/test JDE environments, a **DHCP reservation** gives the benefits of both: the IP is assigned by DHCP but is always the same, bound to the server's MAC address.

```
DHCP Reservation entry (configured on the DHCP server):
  MAC Address:   00:50:56:AB:12:34   (the server's NIC MAC)
  Reserved IP:   10.10.5.20
  Hostname:      jde-dev-es01

Effect: Every time the server requests an IP via DHCP,
        it always gets 10.10.5.20.
```

> ***CNC Rule for production:*** *Never use DHCP reservations for production JDE servers. Use full static IP configuration. For dev/test, reservations are acceptable but static is still preferred.*

Quick Reference: JDE CNC Network Troubleshooting Commands

```
# ── Connectivity Tests ──────────────────────────────────────────────

# Ping a JDE server (Layer 3 check)
ping jde-enterprise01
ping 10.10.1.20

# Test a specific port (Layer 4 check) — Windows
Test-NetConnection -ComputerName jde-enterprise01 -Port 6009
telnet jde-enterprise01 6009

# Test a specific port — Linux
nc -zv jde-enterprise01 6009
# or
bash -c "echo > /dev/tcp/jde-enterprise01/6009" && echo "Port open"

# ── DNS Resolution ──────────────────────────────────────────────────

# Forward lookup (hostname → IP)
nslookup jde-enterprise01

# Reverse lookup (IP → hostname)
nslookup 10.10.1.20

# Flush DNS cache (Windows) — if hostname just changed in DNS
ipconfig /flushdns

# ── Port and Process Checks (run on the JDE server itself) ──────────

# What ports are listening? (Windows)
netstat -an | findstr LISTENING
netstat -an | findstr "6009"

# What ports are listening? (Linux)
ss -tlnp
netstat -tlnp | grep java   # JDE processes are Java

# How many ESTABLISHED connections on JDENET port?
netstat -an | findstr "6009" | findstr ESTABLISHED | Measure-Object -Line

# ── IP and Network Config ───────────────────────────────────────────

# Show IP config (Windows)
ipconfig /all

# Show IP config (Linux)
ip addr show
ip route show

# Show route to a destination
tracert jde-enterprise01     # Windows
traceroute jde-enterprise01  # Linux

# ── JDE Specific ────────────────────────────────────────────────────

# Check jde.ini for server/port config (Windows Deployment Server)
type C:\JDEdwards\E920\ini\JDE.INI | findstr /i "port\|server\|host"

# Check JDE kernel log for connection errors (Linux Enterprise Server)
tail -200 /JDEdwards/E920/logs/jde.log | grep -i "error\|connect\|port"
```

Summary: The 6 Basics and Their CNC Relevance

Topic	Why It Matters for JDE CNC
OSI / TCP/IP	Gives you a systematic framework to diagnose any JDE connectivity problem from bottom up
IP Addressing & Subnets	Every server, firewall rule, and jde.ini entry is based on IPs — you must read and plan them confidently
DNS & Hostnames	JDE uses hostnames everywhere — a wrong DNS record silently breaks inter-component communication
Ports & Protocols	JDE has a specific port map — knowing it lets you write precise firewall rules and verify services instantly
Firewalls & ACLs	You will submit and verify firewall change requests for every JDE deployment and troubleshoot blocked ports
DHCP vs Static IP	All production JDE servers need static IPs — misconfigured DHCP causes catastrophic outages

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